

BTP

PRESENTATION

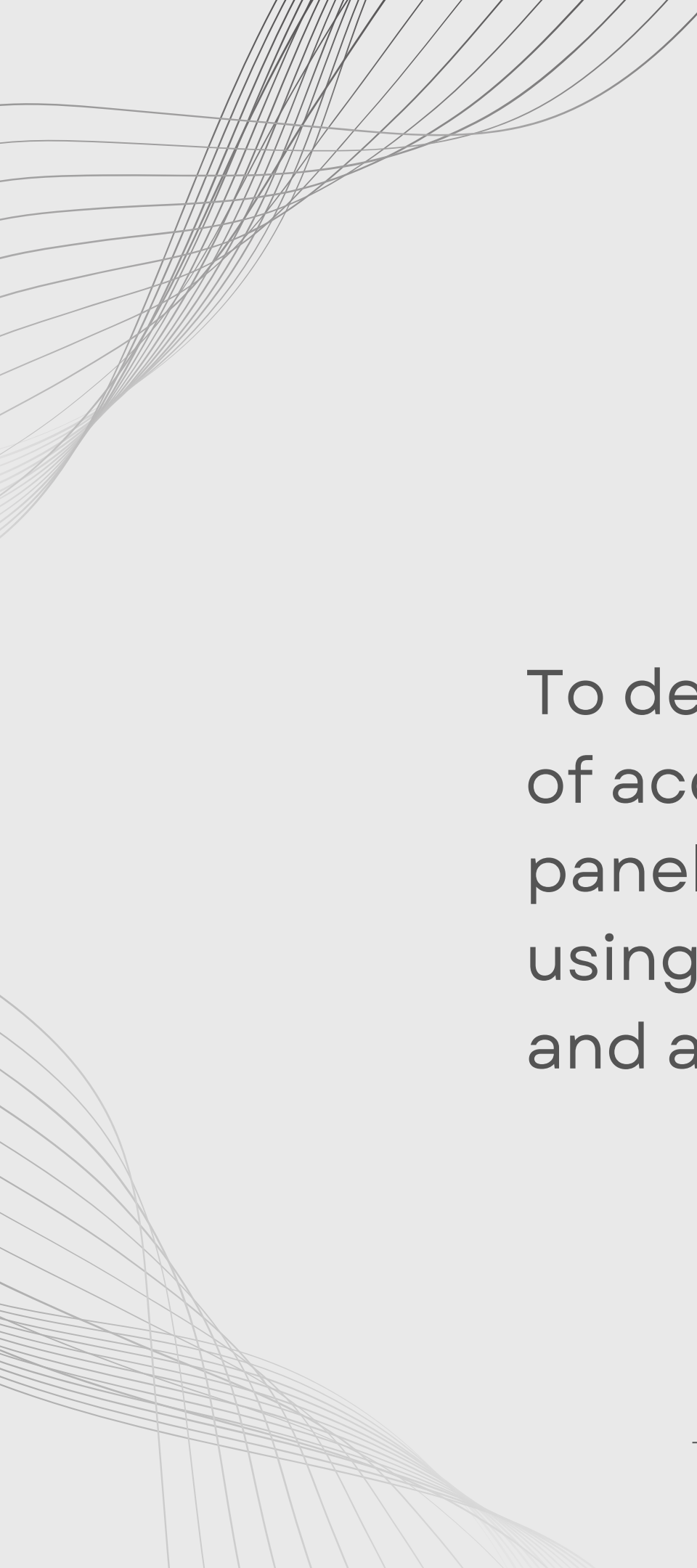
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PROJECT GOAL

To develop an end-to-end regression model capable of accurately predicting the optimal number of solar panels that can be installed on building rooftops using satellite imagery, thereby enabling scalable and automated solar potential assessment.

METHODOLOGY OVERVIEW:

- **Rooftop Detection and Segmentation:**
 - Employed an HRNet + OCR-based architecture to segment all rooftop regions from satellite imagery, creating segment masks.
 - Developed a custom U-Net-based model for identifying masks corresponding to tilted rooftops, enabling the distinction between flat and inclined structures.
 - **Usable Area Estimation:**
 - Applied connected component analysis in conjunction with non-maximum suppression to isolate and refine usable rooftop areas derived from both mask types.
 - Calculated the effective area available for solar panel placement by removing obstructions and considering rooftop geometry.
 - **Solar Panel Placement Modeling:**
 - Determined the maximum number of solar panels that could be placed on the extracted usable areas based on standard panel dimensions.
 - Designed and trained a CNN-based regression model to directly predict the number of solar panels for both flat and tilted rooftops, enabling rapid inference without intermediate area computation.
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Standard Rooftop Detection and Segmentation:

- **Model Architecture:** HRNet with OCR head for high-resolution semantic segmentation.
 - **Training Strategy:** pretrained on the INRIA Aerial Image Labelling dataset, then fine-tuned on the Indian rooftop dataset.
 - **Output:** Binary rooftop masks extracted from satellite imagery for subsequent processing.
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Results of Standard Rooftop Detection and Segmentation

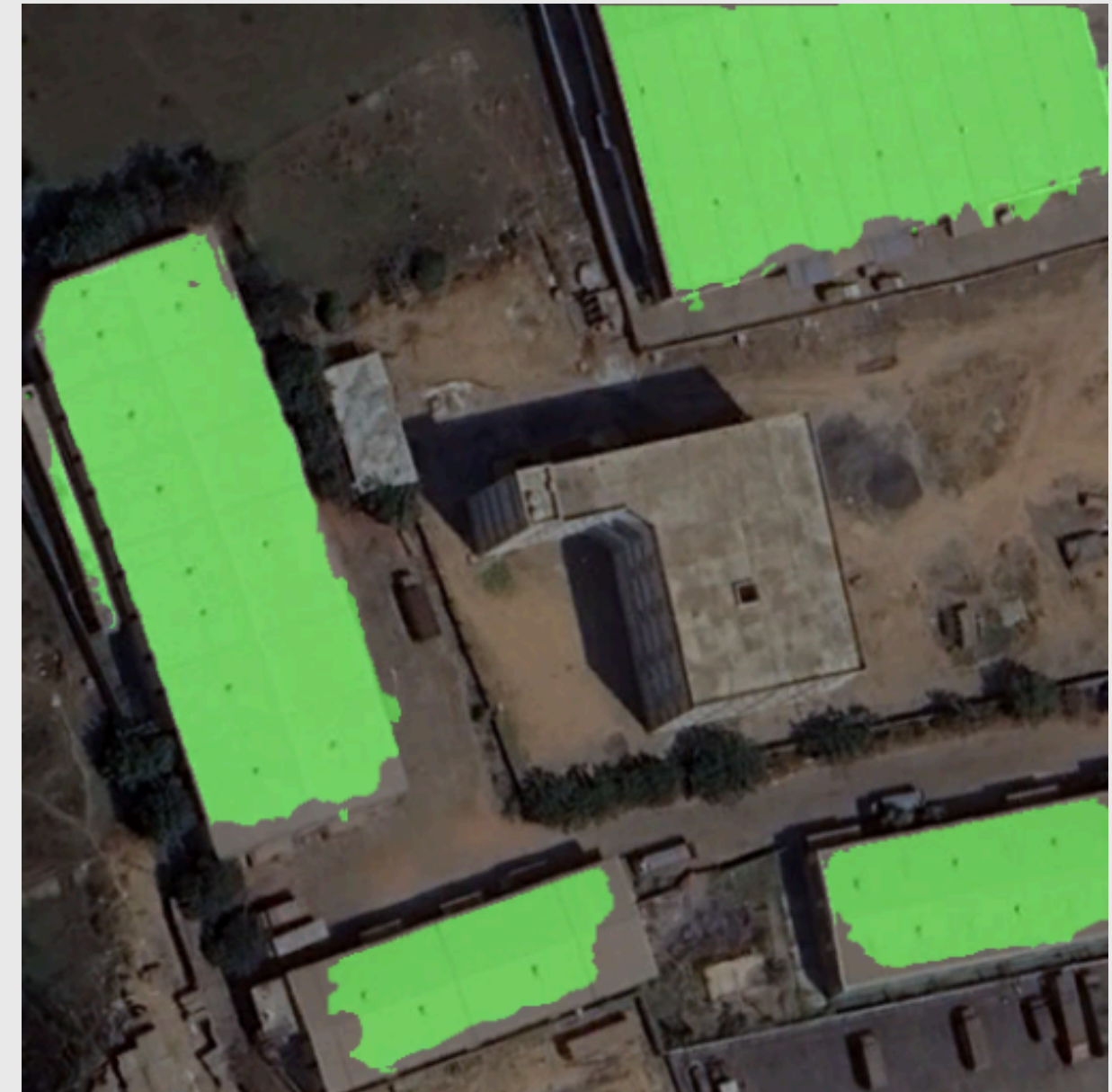




Tilted Rooftop Detection and Segmentation:

- **Model Architecture:** Custom U-Net encoder-decoder with skip connections for tilted roof segmentation
 - **Dataset Creation:** Manually annotated 90 images with tilted roof labels using PixLab polygon tool (includes augmented samples)
 - **Output:** Binary segmentation masks identifying tilted rooftop regions for mask fusion
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Results of Tilted Rooftop Detection and Segmentation



Usable Area Calculation and Mask Fusion

- **Fusion Logic:**

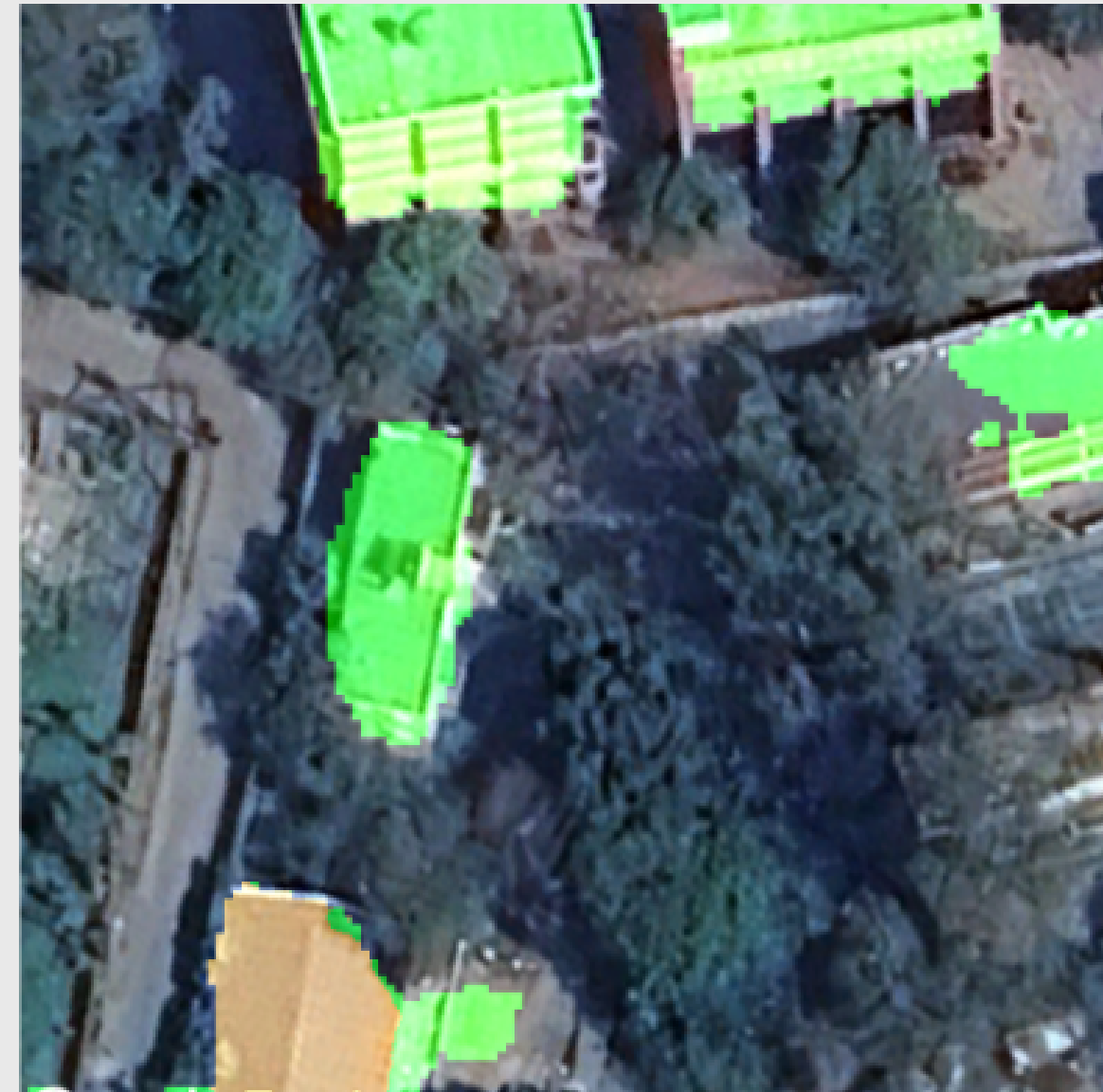
(B -> standard rooftop mask, T -> tilted rooftop mask)

- Fully usable mask: $B \setminus (B \cap T) \rightarrow 100\%$ usable
- Tilt-overlap mask: $B \cap T \rightarrow 70\%$ usable ($\alpha = 0.7$)
- Tilted-only regions discarded as false positives

- **Area Computation:** Connected components analysis on base mask identifies individual rooftops; areas converted to m^2 via ground sampling distance

- **Panel Count Conversion:** Areas divided by panel size (1.7 m^2) with usability factor applied

Results of combined masks



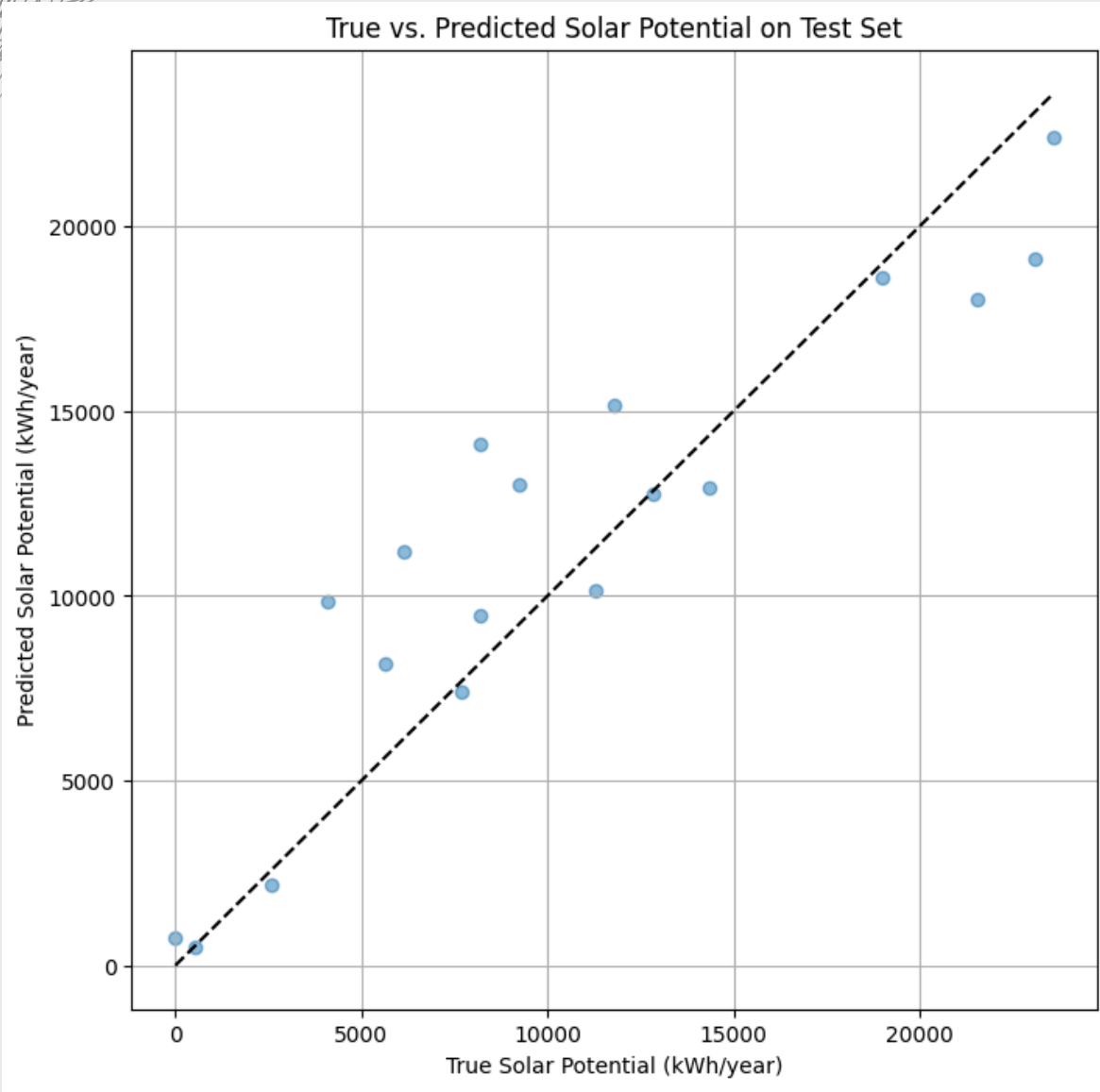
Green → Completely Usable area
Orange → Partialy Usable area (Tilted roof)



End-to-End CNN Regressor for Panel Count Prediction

- **Architecture:** Compact 3-block CNN with dropout for regularization
 - **Input:** Raw satellite imagery
 - **Output:** Two-target regression predicting panel counts (normal & tilted)
 - **Training:** MSE loss on both outputs; Adam optimizer; 50 epochs with batch size 32
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Results of CNN Regressor for Prediction



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Evaluating model on the test set...

--- Test Set Evaluation ---
Root Mean Squared Error (RMSE): 2983.7860
Mean Absolute Error (MAE): 2262.7373
R-squared (R2): 0.8263
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```

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*** Panel Count Prediction Results
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Target	R ²	RMSE	MAE	Acc@±2	Mean Error
Normal panels	0.876	3.62	2.48	87.9%	-0.12
Tilted panels	0.742	2.31	1.47	83.6%	0.08



THANK YOU